Max Profit With K Transactions

AlgoExpert · Very Hard · Dynamic Programming

The problem

- Daily prices, at most k buy/sell pairs (no overlap); max profit.

Intuition

- For each transaction count, track the best profit by day using the best buy so far.

Brute force vs optimal

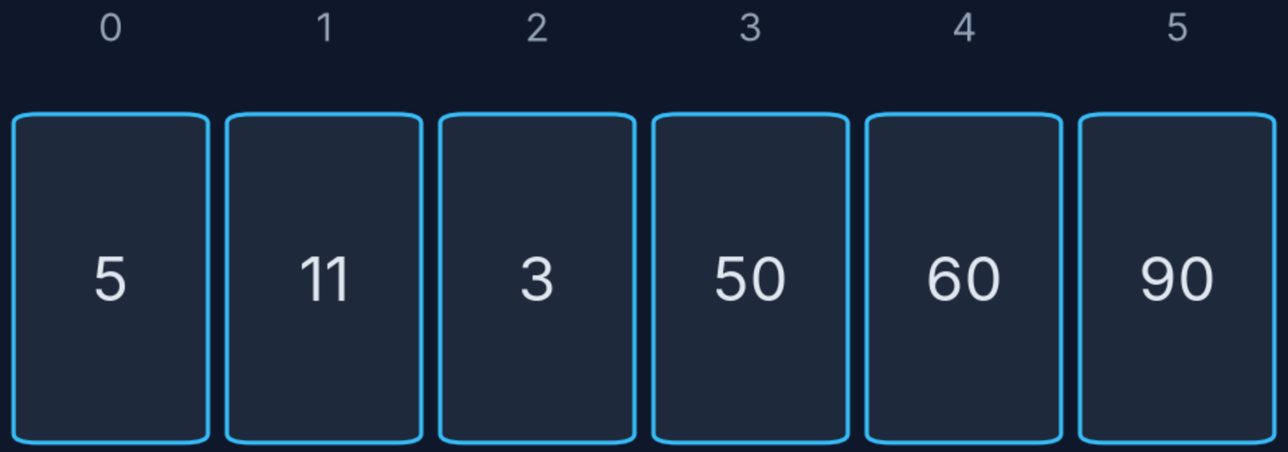
- Try all buy/sell combos → exponential
- DP[t][d] with inner loop → $O(k \cdot n^2)$
- Track best buy → $O(k \cdot n)$, two rows

Approach

- 1. Rows for transactions, columns for days
- 2. Carry max of previous row minus price
- 3. Profit is max of skip or sell today
- 4. Two rows suffice

Worked example

prices



Prices; k = 2

Worked example

	5	11	3	50	60	90
0 tx	0	0	0	0	0	0
1 tx	0	6	6	47	57	87

```
best with 1 transactions: 87
```

1 transaction(s): 87

Worked example

		5	11	3	50	60	90
0 tx	0	0	0	0	0	0	
1 tx	0	6	6	47	57	87	
2 tx	0	6	6	53	63	93	

```
best with 2 transactions: 93
```

2 transaction(s): 93

Java solution

```java
public class MaxProfitWithKTransactions {
    public static int maxProfitWithKTransactions(int[] prices, int k) {
        if (prices.length == 0) return 0;
        int[] prev = new int[prices.length], cur = new int[prices.length];
        for (int t = 1; t <= k; t++) {
            int bestBuy = Integer.MIN_VALUE;
            for (int d = 0; d < prices.length; d++) {
                bestBuy = Math.max(bestBuy, prev[d] - prices[d]);
                cur[d] = Math.max(d > 0 ? cur[d - 1] : 0, bestBuy + prices[d]);
            }
            int[] tmp = prev; prev = cur; cur = tmp;
        }
        return prev[prices.length - 1];
    }
}
```

Complexity

- Time: O(n*k)
- Space: O(n)

Edge cases & pitfalls

- K ≥ n/2 → unlimited transactions
- Empty prices → 0
- Rolling rows

Interview tips & follow-ups

- Explain bestBuy

Recap

- For each transaction count, track the best profit by day using the best buy so far.
- Complexity: O(n*k)